Differentiation Rules Practice

Power, product, and quotient rules for derivatives

Directions:

- Decide which rule the expression calls for *before* you differentiate. A sum of powers needs only the power rule; a product of two non-constant factors needs the product rule; one expression divided by another needs the quotient rule.
- Rewrite negative and fractional powers first: $\dfrac{5}{x} = 5x^{-1}$ and $\sqrt{x} = x^{1/2}$.
- Simplify your answer, and keep a quotient-rule result over a single squared denominator.

1. Differentiate: $y = 4x^5 - 3x^2 + 7$

Answer: ____________________

2. Differentiate: $y = 2x^3 + \dfrac{5}{x}$ (rewrite the second term as a power first)

Answer: ____________________

3. Differentiate: $y = (x^2 + 1)\,(3x - 4)$ (product rule; then expand and collect)

Answer: ____________________

4. Differentiate: $y = \dfrac{2x + 1}{x - 3}$ (quotient rule; leave the denominator squared)

Answer: ____________________

5. Differentiate: $y = \sqrt{x} + \dfrac{6}{x^3}$

Answer: ____________________

6. Differentiate: $y = (3x^2 - 1)\,(x^3 + 2x)$ (product rule, then collect like terms)

Answer: ________________

7. Differentiate: $y = \dfrac{x^2}{x^2 + 1}$ (quotient rule; the numerator simplifies a long way)

Answer: ________________

8. Let $f(x) = (x + 2)\,(x^2 - 5)$.

(a) Find $f'(x)$: ________

(b) Find the slope of the tangent line at $x = 2$: ________

9. Let $g(x) = \dfrac{2x-1}{x^2+3}$.

(a) Find $g'(x)$, over a single squared denominator: _________

(b) Find $g'(1)$ as an exact fraction: _________

10. **Challenge.** Let $f(x) = \dfrac{x^2+3x}{x-1}$.

(a) Find $f'(x)$ and simplify the numerator completely: _________

(b) Evaluate $f'(3)$: _________

(c) Explain what your answer to (b) says about the graph of f at $x = 3$.